AWS SAA-C03 60-Day Challenge - Day 1 to Day 4 Full Tutorial

By Solomon Igein - From Zero to CloudWatch + Snapshot

Region: us-east-1 (N. Virginia) | **Instance:** day2-test-server i-0cae82d2c78d9694d

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Day 1: Launch Your First EC2 Server

You should see `active (running)` in green after this.

1. Go to **AWS Console -> EC2 -> Launch instance**
2. Name: `day2-test-server`, AMI: **Amazon Linux 2023**, Type: `t3.micro` (Free tier)
3. Key pair: **Proceed without a key pair** - we will use Session Manager (more secure for SAA-C03)
4. Network: **Create security group** -> Allow SSH from My IP only for now
5. Storage: `8 GiB gp3` (default)
6. User data - paste this to install Apache:

bash

```bash
#!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo "Hello from SAA-C03 Day 1 - $(hostname -f)" > /var/www/html/inde
```

7. Click **Launch instance**

8. Wait 30 sec -> Go to EC2 -> Instances -> you should see `day2-test-server` -> State: `running` in green.

SAA-C03 concept: EC2 is a virtual server in AWS. t3.micro is burstable, free tier eligible.

Day 2: Security Hardening with IAM Roles + SSM

You should see `Connected` in Session Manager, no SSH key needed.

1. Go to **IAM -> Roles -> Create role** -> Trusted entity: **AWS Service** -> Use case: **EC2** -> Next
2. Add policies: Search `AmazonSSMManagedInstanceCore` -> Check it -> Search `CloudWatchAgentServerPolicy` -> Check it -> Next -> Role name: `EC2-SSM-CW-Role` -> Create role
3. Go to **EC2 -> Instances -> i-0cae82d2c78d9694d -> Actions -> Security -> Modify IAM role** -> Select `EC2-SSM-CW-Role` -> Update IAM role
4. Go to **EC2 -> Instances -> Select instance -> Connect -> Session Manager tab -> Connect**
5. You should see a black terminal with `sh-5.2$` prompt. If you see `Connection failed`, wait 2 min for role to attach.

Then test:

```bash
whoami
sudo yum update -y
```

SAA-C03 concept: Never use SSH keys if you can use SSM. IAM Roles give EC2 permissions to talk to AWS services securely, no hardcoded keys.

Day 3: CloudWatch Basics - Why Default Metrics Are Not Enough

You should see `CPUUtilization` graph only, no RAM or Disk.

1. Go to **CloudWatch -> Metrics -> All metrics -> EC2 -> Per-Instance Metrics**
2. Search `i-0cae82d2c78d9694d` -> You will see only `CPUUtilization`, `StatusCheckFailed`, `NetworkIn` etc.
3. Question: Where is RAM? Where is Disk? Answer: EC2 hypervisor cannot see inside OS. You need CloudWatch Agent.
4. Go to **CloudWatch -> Alarms -> Create alarm** -> Select `CPUUtilization > 80%` -> Create billing alarm > $1 to save cost.

SAA-C03 concept: Default EC2 metrics are hypervisor-level. For memory and disk (OS-level), install CWAgent.

Day 4: CloudWatch Agent - The Hard Day (RAM 28.8% + Disk + Snapshot)

This is where most people fail. You had 4 errors, we fixed all.

Step A: Install Agent via SSM (no SSH)

1. In your black Session Manager terminal for `i-0cae82d2c78d9694d`, run:

```bash
sudo yum install -y amazon-cloudwatch-agent
```

2. You may see terminal show `(END)` in green at bottom - This is the pager bug (your H5.png).

Fix: Click inside black window, press `q` (lowercase). If stuck, press `Ctrl+C` then `q`. If still stuck, open new Session Manager tab.

Step B: Fix IAM if agent has no data (Your H4.png issue)

If you go to CloudWatch -> Metrics and see no `CWAgent` namespace after install, you missed `CloudWatchAgentServerPolicy`.

1. Go to **IAM -> Roles -> EC2-SSM-CW-Role -> Add permissions -> Attach policies**
2. Search `CloudWatchAgentServerPolicy` -> Attach. Wait 60 sec.

Step C: Create the Config File (Your H5.png fail to fetch config error)

The error `fail to fetch json config: open /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json: no such file or directory` means config file doesn't exist.

1. Run this in ONE paste (creates folder + file):

bash

```bash
sudo mkdir -p /opt/aws/amazon-cloudwatch-agent/etc && sudo bash -c 'c
{
  "metrics": {
    "metrics_collected": {
      "mem": { "measurement": ["mem_used_percent"], "metrics_collecti
      "disk": { "measurement": ["used_percent"], "metrics_collection_
    }
  }
}
EOF
'
```

2. Verify file exists:

bash

```bash
cat /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json
```

You should see `mem_used_percent` inside.

Step D: Start Agent - You Should See Validation Succeeded (Your H6.png)

1. Run:

bash

```bash
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl
```

2. You should see `Configuration validation second phase succeeded` and `Configuration validation succeeded` in green - THIS IS YOUR H6.png success screenshot. Screenshot it.

Step E: Verify Live Metrics - The Trophy Shot (Your H7.png -> H10.png)

Wait 2 minutes, then:

1. Go to **CloudWatch -> Metrics -> All metrics**

2. You should now see a new tile `CWAgent 2` with `Custom namespaces` - This is VICTORY (Your H7.png). Click it.

3. Click `device, fstype, host, path 1` -> Check `disk_used_percent` OR click `host 1` -> Check `mem_used_percent`

4. Breadcrumb trick: Click top `All > CWAgent` to go back, then open the other metric.

5. Check both boxes, click `Graphed metrics` tab.

6. You should see: `Graphed metrics (2)` with Blue line `mem_used_percent ~28.8%` and Orange line `nvme0n1p1 xfs / disk_used_percent ~27%` - THIS IS YOUR H10.png trophy. Screenshot it now!

Step F: Create EBS Snapshot - Backup Before Auto Scaling (Your H11.png -> H13.png)

1. Go to **EC2 -> Volumes -> Select vol-0f347c9b043a489a9 8 GiB gp3 for day2-test-server**

2. Click **Actions -> Create snapshot**

3. Description: `day4-before-autoscaling` (Your H11.png) -> Do NOT click Add new tag, just click bottom orange **Create snapshot** button

4. Top green banner: `Successfully created snapshot snap-0289891c0e472868e` (Your H12.png)

5. Go to **EC2 -> Snapshots** -> Select your snapshot -> You should see `Full snapshot size: 2.26 GiB`, `Snapshot status: Completed` (Your H13.png). Screenshot it.

Step G: Cost Optimization - Stop Instance (Your H14.png -> H15.png)

1. Go to **EC2 -> Instances -> Select day2-test-server -> Instance state -> Stop instance**

2. Pop-up: `Stop instance` -> `Can stop` -> Click **Stop** (Your H14.png)

3. Green banner: `Successfully initiated stopping of i-0cae82d2c78d9694d` (Your H15.png)

4. State becomes `Stopping` then `Stopped` - You now pay only for 8 GiB EBS ($0.80/month), no CPU.

You are done with Day 4!

Day 4 Proof Checklist For GitHub

- H6.png: Configuration validation succeeded
- H10.png: Graphed metrics (2) - mem_used_percent 28.8% + disk
- H13.png: Snapshot Completed 2.26 GiB snap-0289891c0e472868e
- H15.png: Instance Stopped

Next: Day 5 - Launch Template from snapshot + Auto Scaling Group + ALB